Michael Boiman

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EN 16931	MIT	A2A 1.0
e-invoicing in production	**open-source agent infrastructure**	**agent for this CV**
Peppol BIS 3.0 into SAP ByDesign	github.com/bks-lab/open-bridge	mboiman.bks-lab.com

Quality Engineering | LLM/AI Architecture | CI/CD Automation | Azure & Serverless | Peppol/E-Invoice

Career Profile

Quality Engineer and AI Architect. Quality engineering and AI architecture are not consecutive chapters here, they are two tracks running in parallel and feeding into each other.

In enterprise projects (**DB Vertrieb**, **DVAG**, **TÜV Süd**), quality monitoring systems took shape, AI-driven test automation was added, and legacy migrations were secured with 24/7 validation.

Today that means e-invoicing to EN 16931 in production and a network of agents consulting each other over the open MCP and A2A protocols.

I also pass this knowledge on, in developer and business workshops and in an invited talk at TU Darmstadt, translating hands-on AI for technical and non-technical audiences alike.

PROJECTS & SOLUTIONS

E-Invoicing Automation Platform

Python · Azure Functions · Peppol · Peppol Access Point · SAP ByDesign

Production e-invoicing pipeline compliant with EU standard EN 16931.

- **Problem:** Inbound and outbound invoices in multiple formats (ZUGFeRD, XRechnung, UBL) required EN 16931-compliant processing.
- **Solution:** Peppol connection through a certified access point, hexagonal architecture, automatic transfer into the ERP.
- **Result:** Production system, invoice processing runs automated end-to-end into the ERP. Seven repositories, roughly 2,900 automated test cases, more test code than production code, an enforced coverage threshold per repository.

24/7 Automated Legacy Migration Validator: Dual-Run Quality Engineering

Java · Cucumber · Kibana · Graylog · Elasticsearch

Dual-run validation for safely migrating a mainframe price calculation.

- **Problem:** Millions of price combinations had to migrate without quality loss or data errors.
- **Solution:** 24/7 validation with randomly generated tests, live dashboard and full deviation traceability.
- **Result:** Safe migration without data loss, all combinations validated around the clock.

Technical Skills

Quality Engineering · Specification & Verification Design · Playwright · Behave / pytest (BDD) · Cucumber/Gauge · Python · TypeScript · FastAPI · Claude Code (Skills, Hooks, MCP) · LLMs (Claude, Gemini, GPT) · Agent Orchestration (MCP & A2A) · a2a-sdk & Agent Cards · Google ADK · JSON-RPC 2.0 & SSE · LangChain · Peppol/E-Invoicing (EN 16931) · Azure Functions · Azure DevOps · Azure Monitor & KQL · CI/CD · Docker · Cloudflare (Workers, Pages, R2) · Microsoft 365 / Exchange Online · Grafana · Elasticsearch · Kibana · OpenTelemetry / OTLP · REST APIs · JMeter · Gatling · Agile (Scrum/Kanban)

Education

Diploma in Computer Science (UAS)
Cologne University of Applied Sciences, Gummersbach
2000-2005

ISTQB Certified Tester

2005

Certified Dynatrace Diagnostics Basic Training

2007

Languages

German	Native
English	Fluent (C1)

Agent mesh and MCP-to-A2A gateway

Python · a2a-sdk · A2A Protocol 1.0 · Agent Cards · JSON-RPC 2.0

Three agents, one translator, one open protocol. The agent on my CV page is one of them.

- **Problem:** An agent that only holds its own knowledge guesses at everything outside it.
- **Solution:** Three agents, each with its own subject, consult each other over the open A2A protocol, read-only. In front of them a stateless gateway offering the same agents to MCP clients as three tools.
- **Result:** Publicly retrievable agent card, 98 automated test cases, generic framework open under MIT.

Quality Dashboard · Real-time Overview

Grafana · Elasticsearch · Python · CI/CD · Playwright

Real-time quality overview for enterprise go/no-go decisions.

- **Problem:** Release decisions required hours of gathering data from the scattered test tools of several teams and environments.
- **Solution:** End-to-end pipeline with live dashboard, multi-source integration and automated PDF reports.
- **Result:** Data-driven release decisions in real time instead of manual collection.

AI-driven Test Automation

Python · OpenAI GPT-4 · Playwright · IDE Integration · Jira API

Guideline-based AI agent in the IDE that generates tests from Jira stories.

- **Problem:** Deriving test cases from a story is manual work, and a model without constraints invents steps and selectors the application does not have.
- **Solution:** The agent reads the story over the Jira API; the project's own conventions (page objects, naming rules, permitted selectors) sit beside it as a rule file and bound what it may produce.
- **Result:** The test comes out in the style of the existing Playwright project. Typing an empty file turns into reviewing a draft.

AI Context Orchestrator: Multi-Repo Development Platform

Python · YAML · Claude Code Plugins · Hooks API · GitHub Actions

Central control system for AI-assisted development across customer, partner and internal repos.

- **Problem:** Development across many repos and multiple customers required constant context switching and manual coordination.
- **Solution:** Orchestration hub with a three-way split registry, one skill tree for every tool and a hook-based permission system.
- **Result:** Workflows from incident response to meeting documentation largely automated.

Appointment enquiry over WhatsApp and phone

Python · FastAPI · WhatsApp Business Cloud API · Speech-to-Text · Text-to-Speech

Two channels on one domain core, WhatsApp and phone, reading free slots live from the booking portal.

- **Problem:** Calls were lost during work, and a language model must never invent an appointment.
- **Solution:** The model runs the conversation, the free slots come live from the portal API. It pencils an appointment in but never books it.
- **Result:** The WhatsApp channel has been in production at the client since 08/2026 with real end users and data residency in Germany; consent comes before the first substantive sentence.

PROFESSIONAL EXPERIENCE

LLM Infrastructure Architect & Automation Engineer

since 06/2025

BKS · AI Research & Development

Development of LLM-orchestrated Enterprise Development Ecosystem: Fully integrated system for knowledge management, project automation, and development workflows with Claude, Gemini, and OpenAI GPT.

Key Responsibilities

- **Open-Source Agent Infrastructure**: open-bridge under the MIT licence, public at github.com/bks-lab/open-bridge, with a live demo site. The generic CORE layer (agent runtime, MCP gateway, skill tree, promote flow with a content-leak gate in CI) grows out of my own instance and is published upward by scope, with its own test suite and continuous releases since June 2026. The counts live in the repo, so they cannot go stale here. Attack fixtures for path traversal, secrets and malformed overlays run as their own CI stage
- **Data separation as architecture, not as agreement**: separate instances per client, a rule forbidding access between instances, and a fail-closed contract for materialising shared configuration (the sync aborts rather than writing on doubt). Protection against data leakage at three points: pre-commit hook, pre-push hook and three validators in CI. Plus an adversarial simulation that attacks the protection itself on a schedule, as its own CI job in a sealed sandbox
- **A2A Agent Network**: an a2a-sdk server with a local claude backend, running on a runtime adopted from open-bridge. Three agents (person, company, OSS project) consult each other on read-only edges, fronted by a stateless MCP-to-A2A gateway so MCP-only clients can reach them. Agent Cards at protocol version 1.0, JSON-RPC 2.0. Around 4,600 lines of Python with 98 automated test cases, built as a team effort (four further contributors)
- **OpenTelemetry-based telemetry**: logs, metrics and traces over OTLP into a managed sink, with a fixed field convention and naming scheme, a capped ingest key and read rights separated per client (measured as an access cut, not documented as an intention). Honest maturity: the log path is in operation, metrics and traces are wired and not yet accepted

Tools & Technologies

- LLM Orchestration: Claude (primary), Google Gemini, OpenAI GPT, Model Context Protocol (MCP), Google Agent Development Kit (ADK)
- AI Development Platform: Claude Code, Plugin Architecture (Skills, Hooks, Commands, Agents), YAML/Markdown Configuration
- Multi-Agent: A2A Protocol 1.0, a2a-sdk, Agent Cards, MCP-to-A2A gateway, JSON-RPC 2.0, SSE Streaming
- Backend: Python 3.13, FastAPI, uvx Distribution, Git Submodules
- Automation: GitHub Actions, GitHub GraphQL API, gh CLI
- Integration: SharePoint API, Microsoft Graph, Exchange Online, Elasticsearch, Kibana, OpenTelemetry/OTLP
- Betrieb: Cloudflare (Workers, Pages, R2, DNS), Tailscale, launchd, restic/rclone
- Development: Bash Scripting, jq, Docker, Multi-Agent Systems; orchestrated repos span Python, TypeScript, Rust and Go

Senior Quality Engineer · Test Automation and Monitoring

since 09/2025

TÜV Süd

Quality Engineering for an enterprise platform for AI-powered document processing in medical device certification (MDR/IVDR) with AI chat integration and data pipeline automation.

Key Responsibilities

- **BDD API Test Framework**: Development of comprehensive test framework with Python/Behave, self-healing authentication, HTML report generator with embedded API responses
- **E2E Test Automation**: Playwright E2E test suite for complete user journey (Setup -> Upload -> Transformation -> Download) with 500 error handling and route mocking
- **Performance & Monitoring**: Development of Azure Monitor Workbooks (QA Live Testing Companion, Error Investigation Assistant) with KQL for end-to-end monitoring
- **Bug Analysis & Quality Intelligence**: Implementation of comprehensive bug tracking system with WIQL queries, automated dashboards, and executive PDF reporting

Tools & Technologies

- Testing: Behave, Playwright, TypeScript, Python, pytest, Page Object Model, Fixtures
- Monitoring: Azure Monitor Workbooks, KQL, Application Insights, Log Analytics
- Backend: FastAPI, Python, Pydantic, SQLModel, Azure Functions, Azure Service Bus
- Frontend: React, TypeScript, Vite, TanStack Router
- AI Integration: Azure OpenAI (GPT-4o), LangChain, Azure AI Search, Embeddings
- DevOps: Azure DevOps Pipelines, Git, Docker, Azure Blob Storage, Poetry

Technical Lead & AI Automation Architect

01/2024-04/2025

BKS

Technical lead & development: Design and implementation of a scalable automation solution for processing e-invoices and PDF invoices fetched from a mailbox of a leading online job platform (under NDA). Full responsibility from requirements analysis through implementation to production go-live.

Focus areas

- **Workflow automation**: Extraction, analysis and classification of incoming invoices using Python and Azure Functions.
- **E-invoice classification & data extraction**: Full processing in accordance with EU standard EN 16931 (ZUGFeRD, XRechnung, Factur-X), including structural and content parsing.
- **Peppol Network Integration**: Connection to European e-invoicing network through a certified Peppol Access Point for automated invoice receipt and dispatch with UBL/CII formats.
- **SAP ByDesign Integration**: Automatic transfer of validated invoice data to SAP ByDesign for seamless ERP integration.

Tools & technologies

- Azure Functions: Serverless orchestration of automation workflows.
- Python: Core logic for data extraction, classification and integration with automated CI/CD processes, packaging, and comprehensive unit and integration tests for quality assurance.
- E-Invoicing: Peppol Network, Access-Point-API, ZUGFeRD/Factur-X, XRechnung, UBL 2.1, CII
- ERP Integration: SAP ByDesign, Salesforce
- Microsoft Graph API: Access to Outlook mailboxes and SharePoint libraries.
- GitHub: Version control, CI/CD pipelines and collaborative development.
- Kibana: Monitoring, log analysis and performance dashboards.
- Streamlit: For developing an interactive web application for visualization.
- AI Automation: Claude AI, Custom Skills Development, Git Analysis, Natural Language Processing
- Protocols: Model Context Protocol (MCP), Agent-to-Agent (A2A), JSON-RPC 2.0

Platform Quality Architect (CI/CD, Monitoring, Automation)

08/2021-05/2025

DVAG

Test management & coordination, as well as introduction of CI/CD with quality-gate setup. Acting as cross-team Quality Lead: onboarding, guidance, consulting and training for the introduction of quality metrics and test automation within the pipeline across teams.

Result: Real-time sprint-quality transparency via Grafana dashboards, quality-gated releases in the teams' CI/CD pipelines: data-driven release decisions replacing manual data gathering.

Tools

- IntelliJ / Visual Studio Code
- Java / JavaScript / TypeScript
- Playwright / Gauge / Karate
- Python for processing and analysis of test and reporting data
- CI/CD / GitHub (reusable workflows)
- Helm / Containers
- JIRA / Confluence
- Grafana
- Behaviour Driven Development
- Consumer-Driven Contracts (Pact.io)

AI-driven Automated QA Environment for Energy Infrastructure

07/2024-01/2025

AkkuSwap Startup

QA Leadership for EU-wide Battery Swap Infrastructure: AI-driven simulation tool for battery swap station network with energy infrastructure integration

Technical Stack

- Infrastructure: Linux, Docker, Azure OpenAI
- Automation: pytest, AI code generation integration patterns
- Monitoring: Grafana, Azure Monitoring
- Energy Systems: Battery swap infrastructure, grid integration simulation

Key Achievements

- Established QA methodology for AI-driven infrastructure simulation
- Created testing framework for energy sector critical systems
- Validated proof-of-concept with measurable reliability improvements

AI Engineering Lead & ML Solutions Architect

04/2023-04/2024

BKS on behalf of a client in the sustainability sector

Project management & technical lead: Development and deployment of an AI-driven email bot to automate customer communication, AI-assisted processing of incoming emails and triggering of follow-up activities in existing SAP systems. Responsible for end-to-end project coordination and team management from idea to implementation.

Tools & technologies

- Azure Functions: For serverless application architectures and ML model execution.
- Python: Main programming language for the email bot and ML models.
- Microsoft Graph API: For accessing and processing emails in Microsoft Outlook 365.
- SAP API: For querying customer data and communicating with SAP systems.
- GitHub: For version control, CI/CD pipelines and collaborative development.
- Kibana: For real-time monitoring and system performance analysis.
- Streamlit: For developing an interactive web application for system monitoring.

DevOps Quality Lead & Dashboard Architect

01/2017-05/2021

DB Vertrieb

Test management, test coordination, test automation, agile (Kanban, Scrum, SAFe): Coordination as PO of a cross-team QA team including cross-team onboarding, support, consulting, training

Tools

- IntelliJ
- Java
- Cucumber / Cypress
- JMeter / Gatling
- CI/CD / GitLab CI / Jenkins
- Helm / Docker
- JIRA / Confluence
- Kibana / Grafana / Instana / Graylog
- Behaviour Driven Development
- Consumer Driven Contracts (Spring Cloud Contract)
- Ionic (hybrid app)
- Selenium

Earlier Positions

09/2015-01/2017	**Performance Engineering Lead & Test Architect**, DB Systel	Load and performance tests and analysis (test manager, test designer, analyst)
04/2015-09/2015	**Senior iOS Developer & Mobile Architect**, Telekom	Senior iOS Developer
05/2009-03/2015	**Quality Assurance Lead & Test Automation Architect**, Siemens	QA, test management, test automation, requirements engineering
03/2009-05/2009	**Performance Test Engineer & Load Testing Specialist**, ING-DIBA	Load and performance tester
01/2006-12/2008	**Performance Test Engineer & Quality Consultant**, British Telecom, Mobiliar, DB-Systel, Sparkassen Informatik, Loyalty Partner, Telekom, Itelium, Deutsche Post, Postbank	Load and performance tester / tester

WORKSHOPS & PRESENTATIONS

Guest Lecture: AI-Assisted Software Development in Practice
TU Darmstadt · Information Systems Group · Lab course “AI Startup: From Idea to Execution” · 04/2026

Workshop: Development with Generative Language Models
Developer Workshop · Agent-based Software Development · 06/2025

Presentation: Efficient Documentation through Automation
Enterprise Presentation · AI-powered Documentation Workflows · 04/2025

Applications of AI in Business Context
AI Workshop for Business Integration · 2023